استرس
ثانية ميكا
1

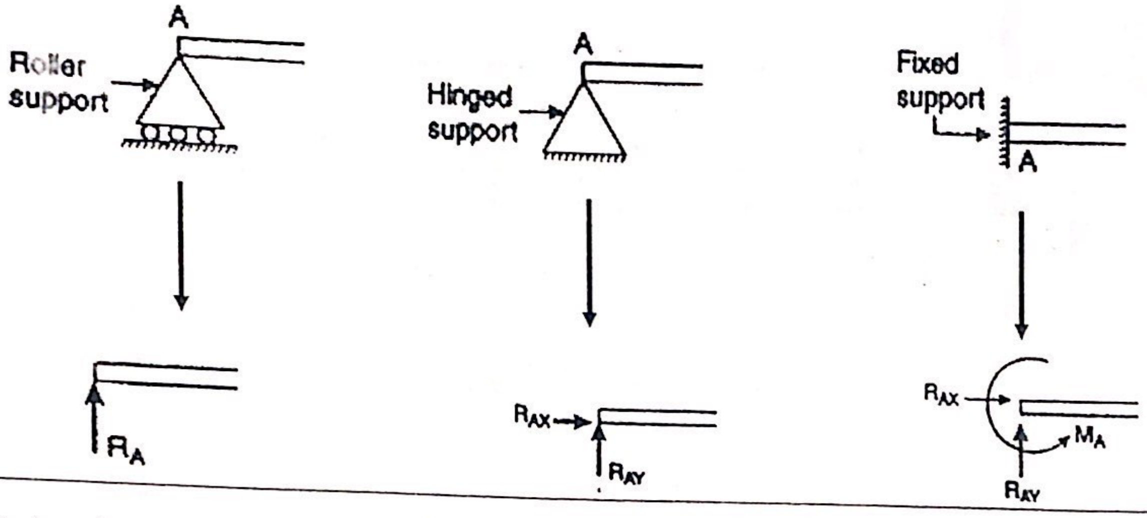
STRESS ANALYSIS

..............................(1)

2^{nd} MECH

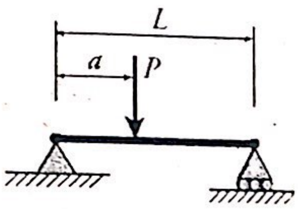
Revision

1. Supports

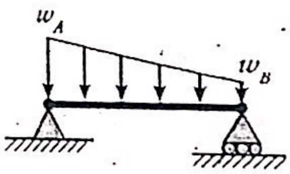


2. Loads

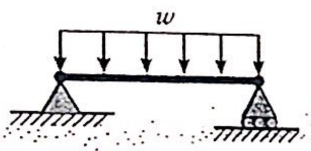
POINT LOAD



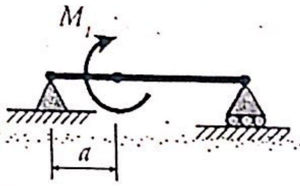
Non-Uniform DISTRIBUTED LOAD



UNIFORMLY DISTRIBUTED LOAD



POINT MOMENT



3. Equilibrium Equations

In This course, the system is always in equilibrium which means that the internal reactions balance the applied external loads. Therefore the sum of components of forces and moments about any axis must be equal zero.

$$\sum F_x = 0 \qquad \sum M_x = 0$$
$$\sum F_y = 0 \qquad \sum M_y = 0$$
$$\sum F_z = 0 \qquad \sum M_z = 0$$

4. Reactions

a) Indicate the unknown reaction components depending on the type of support (hinged, roller or fixation).

b) Replace distributed loads (if any) by its equivalent concentrated load at the CG of the distributed load.

c) Apply the equilibrium equation to find the unknown reactions.

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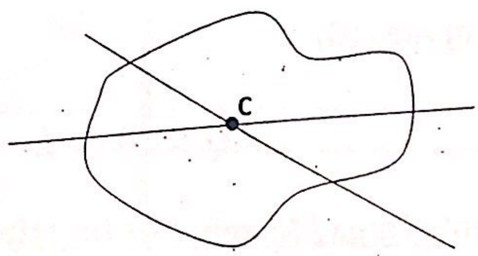
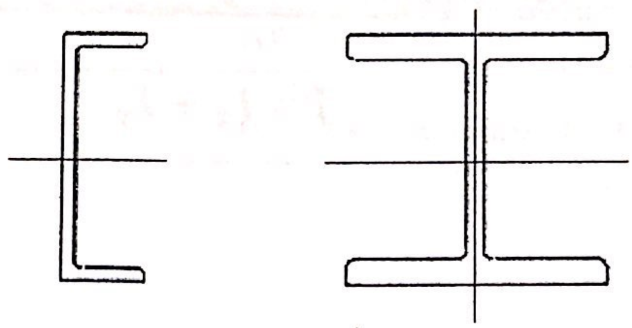
5. Centroid (C)

The centroid of an area is the point at which any line drawn through it in any direction divides the area into two equal areas.



If the area is symmetric around an axis → Its centroid lies on this axis.

Examples:



Procedure to Find Centroid:

a) Divide the given area into a number of shapes such that the centroid of each shape is known.

b) Calculate the of each shape

c) Assume reference axes and calculate the vertical and horizontal distances from each shape centroid to those axes (x_i and y_i).

d) Calculate $\bar{x}$ and $\bar{y}$ using:

$$\bar{x} = \frac{\sum A_i x_i}{\sum A_i} \quad \& \quad \bar{y} = \frac{\sum A_i y_i}{\sum A_i}$$

$\bar{x}$: Distance of centroid from the vertical reference axis

$\bar{y}$: Distance of centroid from the horizontal reference axis

6. First Moment of Area (Q)

$$Q = A \times L$$

7. Second moment of area / Moment of Inertia (I)

a) Regular shapes → Formula

b) Irregular shapes → Parallel axis Theorem (نظرية المحاور المتوازية)

8. Polar Moment of Inertia (J)

$$J = I_x + I_y$$

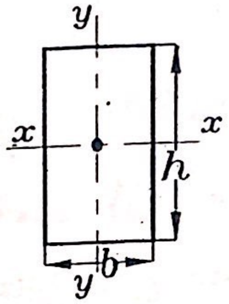
Geometrical Properties of Plane Sections

Rectangle

$A = b \times h$

$I_x = \frac{b \times h^3}{12}$

$I_y = \frac{h \times b^3}{12}$

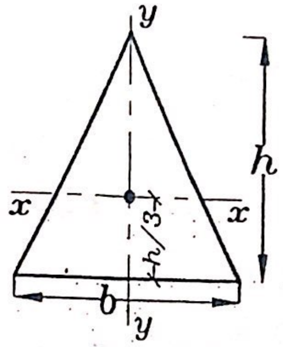


Triangle

$A = \frac{1}{2} \times b \times h$

$I_x = \frac{b \times h^3}{36}$

$I_y = \frac{h \times b^3}{48}$

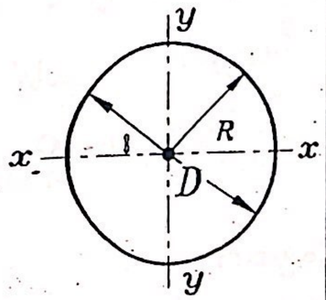


Circle

$A = \frac{\pi D^2}{4}$

$I_x = I_y = I = \frac{\pi D^4}{64}$

$J = \frac{\pi D^4}{32}$

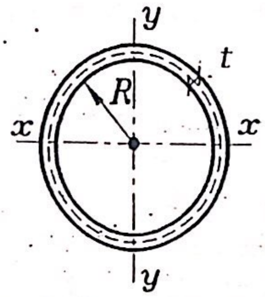


Thin Tube

$A = 2\pi R t = \pi D t$

$J = 2I = 2\pi R^3 t$

$I_x = I_y = I = \pi R^3 t$

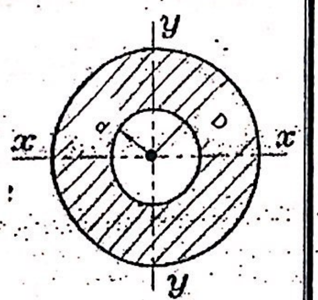


Hollow Circle

$A = \frac{\pi}{4}(D^2 - d^2)$

$I = \frac{\pi}{64}(D^4 - d^4)$

$J = \frac{\pi}{32}(D^4 - d^4)$

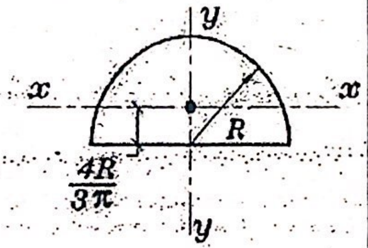


Semi-Circle

$A = \frac{\pi d^2}{8}$

$I_x = 0.11 \times R^4$

$I_y = \frac{\pi}{8} \times R^4 = 0.3926 R^4$



Note:

A: Area

I : Moment of Inertia

J : Polar Moment of Inertia

In General:

$J = I_x + I_y$

For circular cross section $I_x = I_y = I \quad \therefore J = 2I$

* كيفية رسم الـ loading diagrams

① **NFD** Normal Force Diagram

أتحرك بإيدي من الشمال إلى اليمين وأقف عند أي قوة عمودية على القطاع ومجرد إني فقط أتخطاها وأشوف، إذا كانت بتضغط على إيدي (يبقى compression)؛ أتحرك لتحت بقيمة هذه القوة وإذا كانت خارجة من إيدي (يبقى tension)؛ أتحرك لفوق بقيمة هذه القوة.

② **SFD** Shear Force Diagram

بردو متحرك من الشمال إلى اليمين وأطلع مع الطالع وأنزل مع النازل بقيمة القوة.

③ **BMD** Bending Moment Diagram

- هأقف عند أي نقطة من النقاط التالية.

1- Ends of beam (بداية ونهاية الـ beam)
2- Concentrated loads
3- Concentrated Moments → يظهر عندها خط رأسي في الـ BMD
4- Ends of distributed loads (بدايته ونهايته)

- ثم أحسب الـ moment يمين أو شمال هذه النقطة

- وأوقع القيمة المحسوبة ناحية ذيل سهم الـ moment

Type of Stresses

- Mechanical stresses
- Thermal stresses

* Mechanical stresses:

1) Normal stress (σ)
2) Shear stress (τ)

1 Normal stresses: [σ]

- Due to Normal Force

$$\sigma = \pm \frac{F}{A}$$

- Due to Bending moment

$$\sigma = \pm \frac{M_x}{I_x} y \pm \frac{M_y}{I_y} x$$

2 Shear stresses: [τ]

- Direct shear

$$\tau = \frac{F}{A_{sh}}$$

- Transverse shear

$$\tau = \frac{VQ}{It}$$

- Due to Torsion.

$$\tau = \frac{Tr}{J}$$

* Shear stress due to Torsion

$$\tau = \frac{Tr}{J}$$

T : Torque (Nm / Nmm)

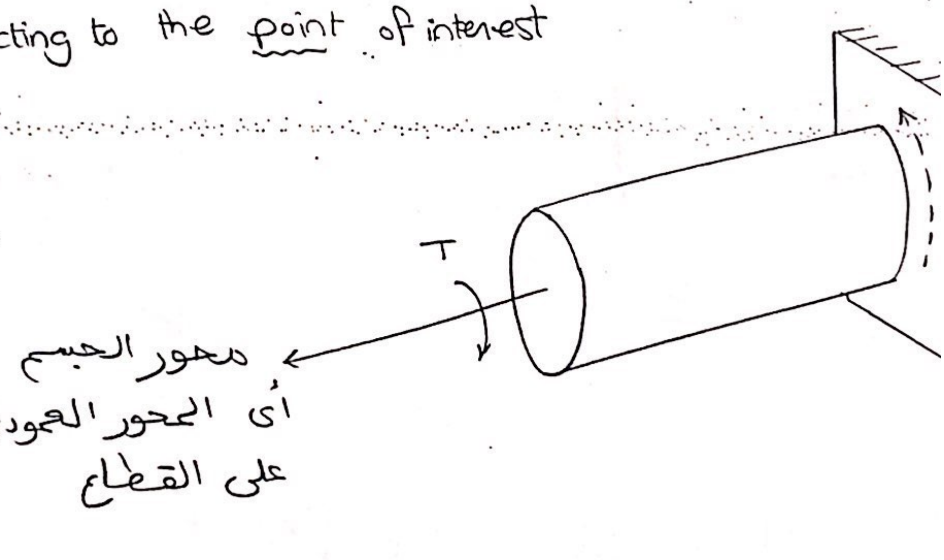
J : polar moment of inertia (mm^4)

In general $J = I_x + I_y$

→ For circular solid cross section $= \frac{\pi d^4}{32}$

→ For circular hollow cross section $= \frac{\pi}{32}\left(d_o^4 - d_i^4\right)$

r : Radial distance from the axis around which the torque is acting to the point of interest



Properties of Area Solved Examples

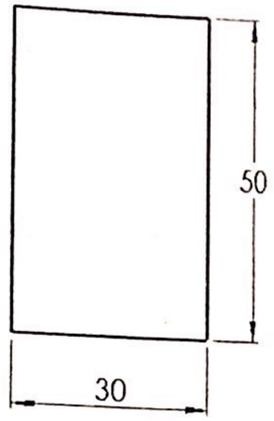
$\bar{x} = 0 \quad \bar{y} = 0$

$A = 1500 \text{ mm}^2$

$I_x = 312500 \text{ mm}^4$

$I_y = 112500 \text{ mm}^4$

$J = 425000 \text{ mm}^4$



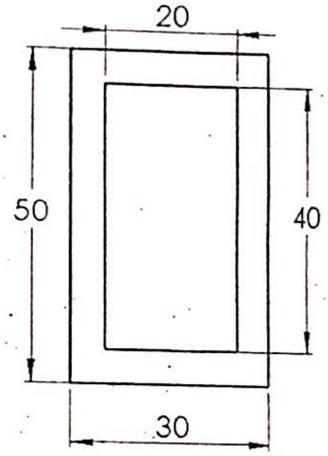
$\bar{x} = 0 \quad \bar{y} = 0$

$A = 700 \text{ mm}^2$

$I_x = 205833.33 \text{ mm}^4$

$I_y = 85833.33 \text{ mm}^4$

$J = 291666.67 \text{ mm}^4$



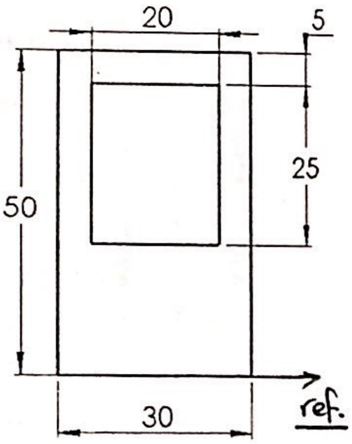
$\bar{x} = 0 \quad \bar{y} = 21.25 \text{ mm}$

$A = 1000 \text{ mm}^2$

$I_x = 244270.83 \text{ mm}^4$

$I_y = 95833.33 \text{ mm}^4$

$J = 340104.17 \text{ mm}^4$



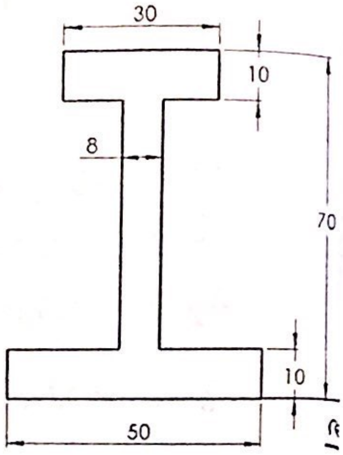
$\bar{x} = 0 \qquad \bar{y} = 30\ mm$

$A = 1200\ mm^2$

$I_x = 780000\ mm^4$

$I_y = 128800\ mm^4$

$J = 908800\ mm^4$



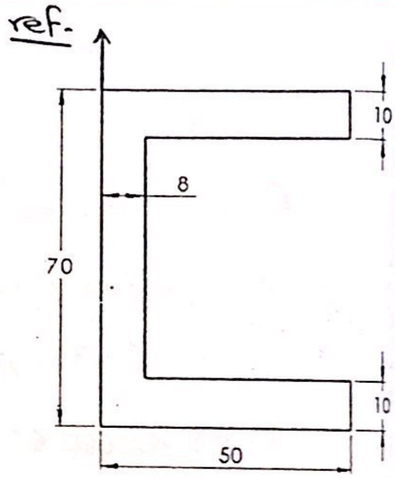
$\bar{x} = 19\ mm \qquad \bar{y} = 0$

$A = 1400\ mm^2$

$I_x = 991666.67\ mm^4$

$I_y = 336466.67\ mm^4$

$J = 1328133.33\ mm^4$



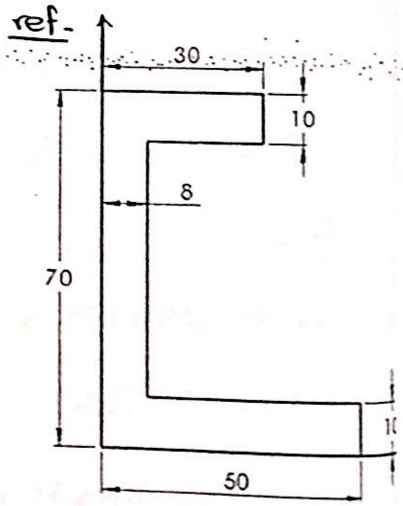
$\bar{x} = 15.5\ mm \qquad \bar{y} = 30\ mm$

$A = 1200\ mm^2$

$I_x = 780000\ mm^4$

$I_y = 226900\ mm^4$

$J = 1006900\ mm^4$



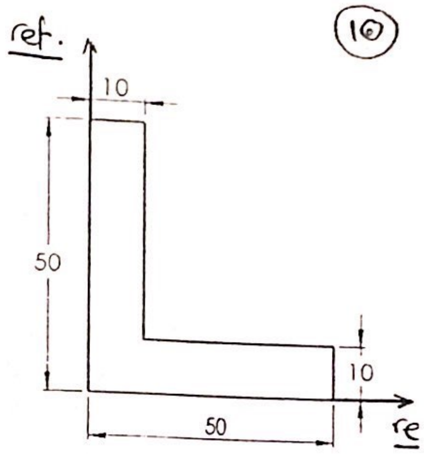
$\bar{x} = 16.11\ mm$ $\bar{y} = 16.11\ mm$

$A = 900\ mm^2$

$I_x = 196388.89\ mm^4$

$I_y = 196388.89\ mm^4$

$J = 392777.78\ mm^4$



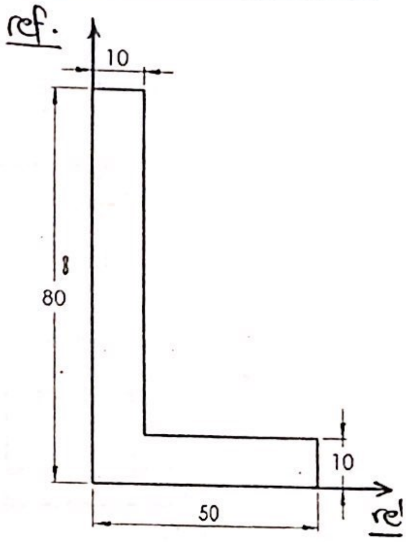
$\bar{x} = 13.33\ mm$ $\bar{y} = 28.33\ mm$

$A = 1200\ mm^2$

$I_x = 756666.67\ mm^4$

$I_y = 226666.67\ mm^4$

$J = 983333.33\ mm^4$



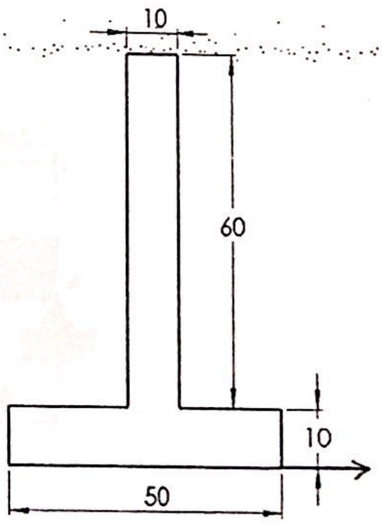
$\bar{x} = 0$ $\bar{y} = 24.1\ mm$

$A = 1100\ mm^2$

$I_x = 518257.58\ mm^4$

$I_y = 109166.67\ mm^4$

$J = 627424.24\ mm^4$

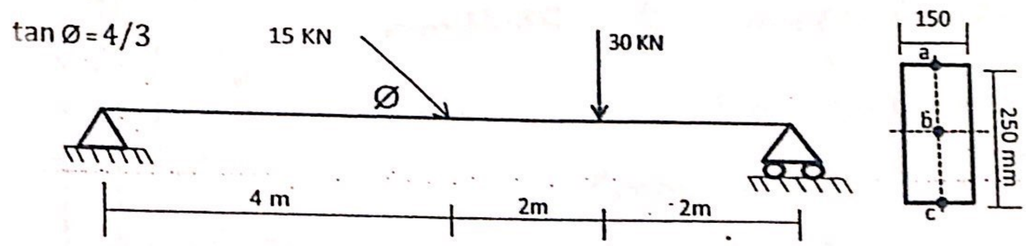


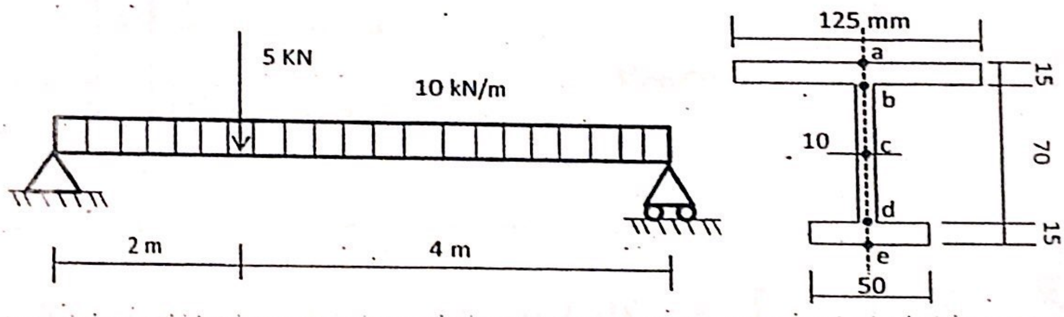
1.

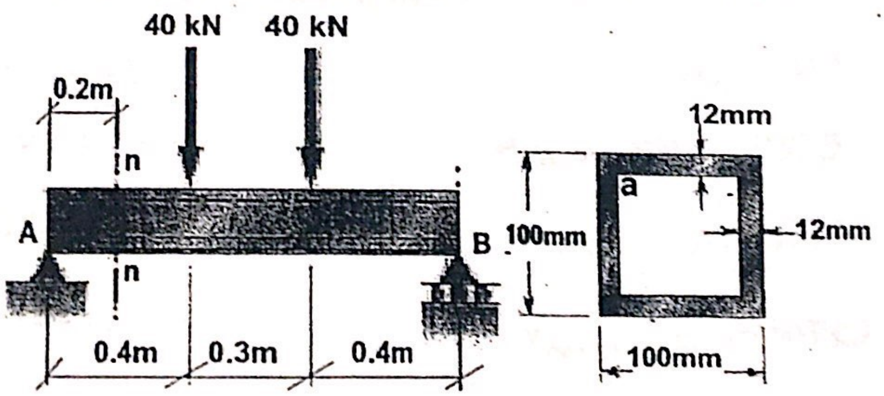
a. **Draw** the normal force, the shearing force, and the bending moment diagrams for each of the following structures.

b. **Calculate** and draw the normal and shear stress diagrams at the critical sections for the given cross sections.

c. **Draw** the state of stress on an infinitesimal element for the shown points **a, b, c, d** and **e.**





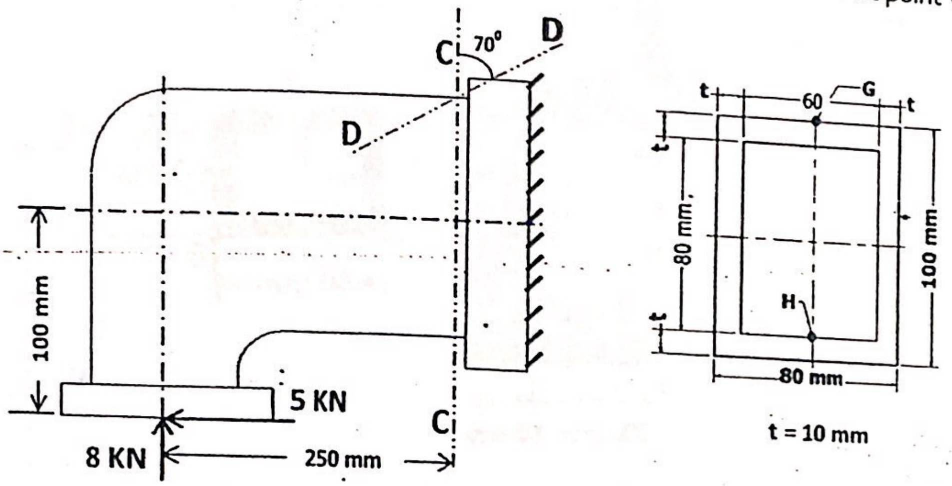


2. The shown frame is subjected to the given loads and has the shown cross section.

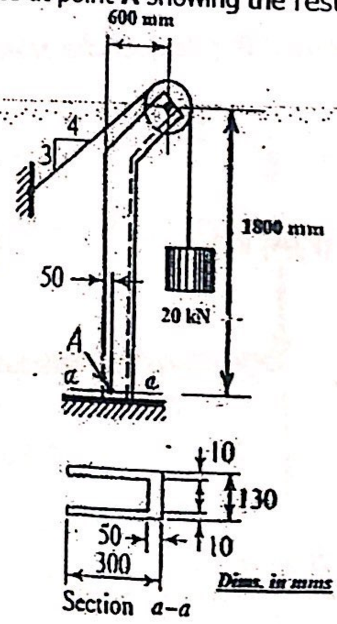
a. **Calculate** and draw the normal and shear stress diagrams at section **C-C**.

b. **Determine** the state of stresses at points **G** and **H** showing the results on infinitesimal elements.

c. **Calculate** the normal and shear stresses on the inclined plane **D-D** at point **G**.

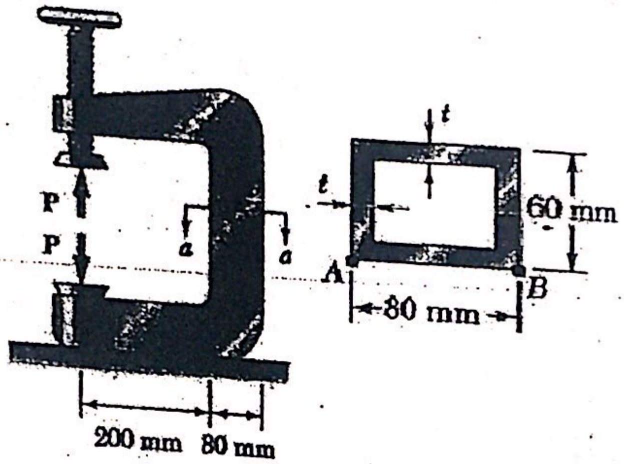


3. A special hoist is loaded with a **20 kN** load suspended by a cable as shown.
Calculate and draw the normal and shear stress diagrams at section **a-a**.
Determine the state of stress at point **A** showing the results on an infinitesimal element.

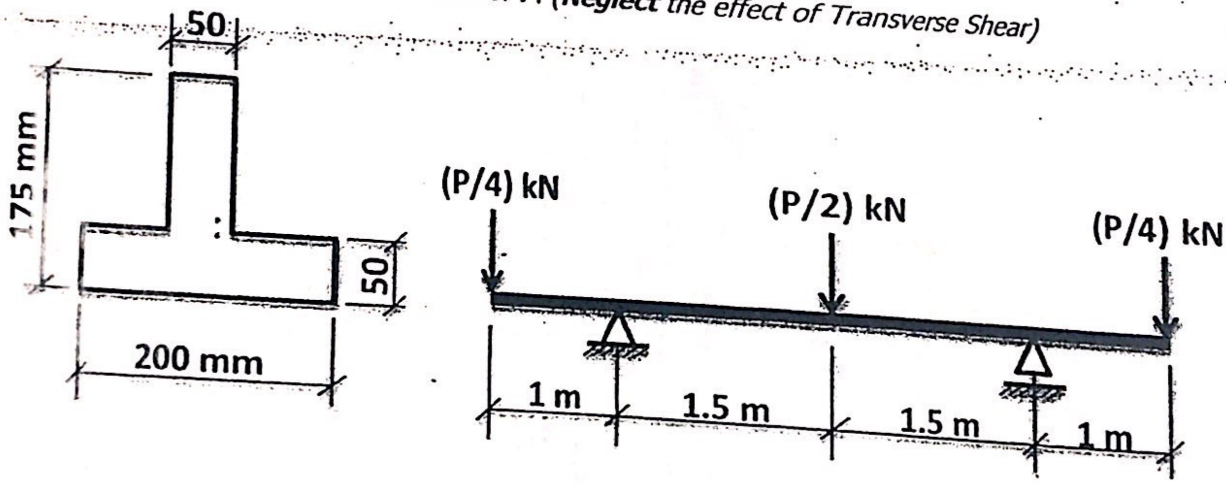


(13)

4. The vertical portion of the press shown in Figure is a rectangular tube having a wall thickness of **t= 10mm.** If the force exerted by the press (**P**) is **30 kN;**
Draw the normal stress distributions in section a-a and determine the maximum tensile and compressive stresses acting in the section **a-a**, Showing the result on infinitesimal element.

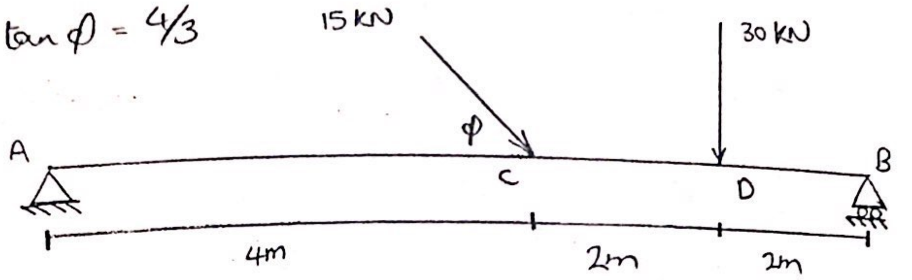


5. The shown beam with overhanging ends is loaded by three concentrated forces. The beam is simply supported and has the shown cross-section .The beam material is grey cast iron having an allowable working stress in tension of **35 MPa** and in compression of **150 MPa.**
Determine the allowable value of **P.** ***(Neglect** the effect of Transverse Shear)*

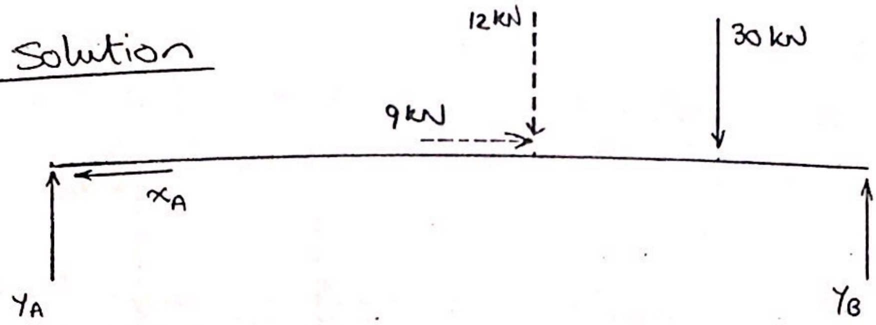


1 a]

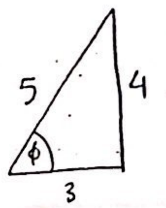
$\tan\phi = 4/3$



Solution



$\tan\phi = 4/3$



$\Rightarrow \cos\phi = 3/5$

$\sin\phi = 4/5$

$\cos\phi = 3/5$

$\sin\phi = 4/5$

$\Rightarrow x_C = 15\cos\phi = \overset{3}{\cancel{15}}\left(\frac{3}{\cancel{5}_1}\right) = 9\ \text{kN}$

$y_C = 15\sin\phi = \overset{3}{\cancel{15}}\left(\frac{4}{\cancel{5}_1}\right) = 12\ \text{kN}$

1) Reactions: x_A, y_A & y_B

we will use equilibrium equations;

$\sum F_x = 0$ [($\rightarrow$) +ve direction]

$-x_A + 9 = 0 \Rightarrow x_A = 9\ \text{kN}$

$x_A = 9\ \text{kN}$

$\sum M @ A = 0$ [($\circlearrowright$) +ve direction]

$$12(4) + 30(6) - Y_B(8) = 0$$

$$\Rightarrow Y_B = 28.5 \text{ kN}$$

$\Sigma F_y = 0$ [(↑) +ve direction]

$$Y_A - 12 - 30 + 28.5 = 0$$

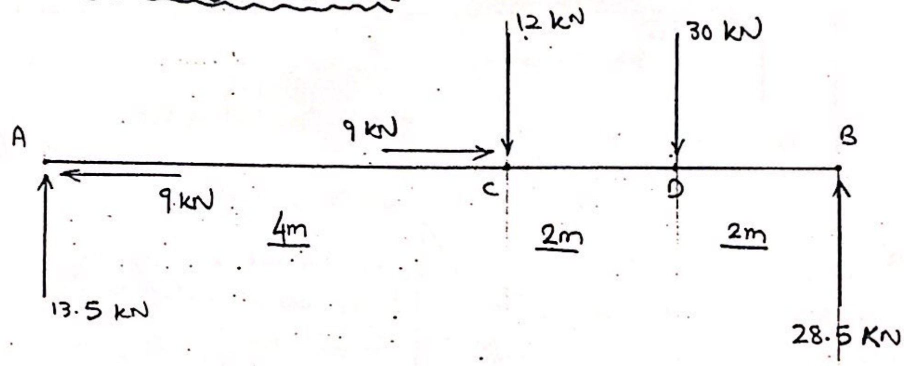
$$\Rightarrow Y_A = 13.5 \text{ kN}$$

2) Loading diagrams :

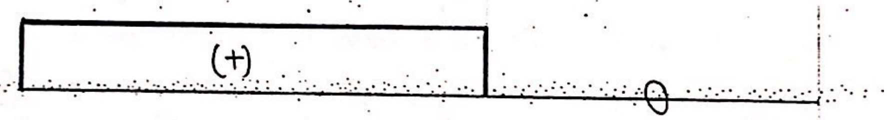
$X_A = 9 \text{kN}$

$Y_A = 13.5 \text{ kN}$

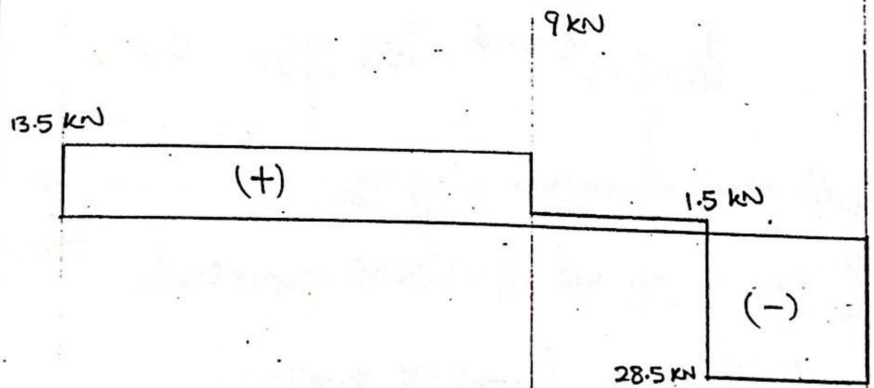
$Y_B = 28.5 \text{ kN}$



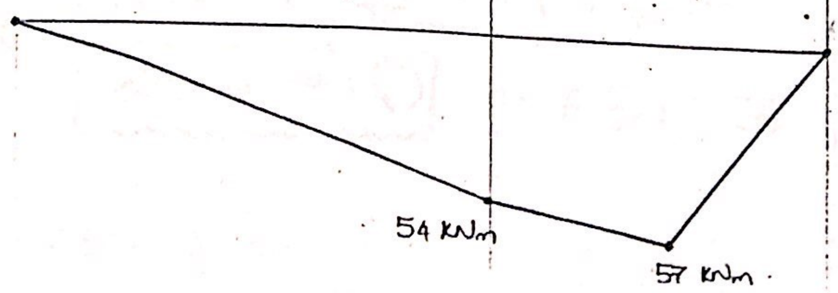
NFD



SFD



BMD



Note:

Critical section is the section exposed to maximum bending moment & zero shear.

أي هو القطاع العرضي لأكبر bending moment ويحدث عنده zero shear.

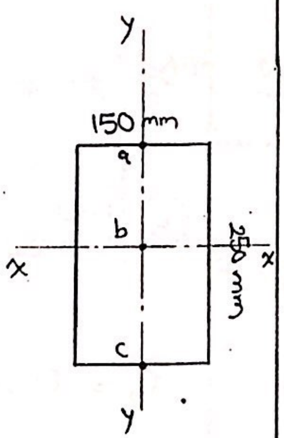
Now, to draw the stress diagrams (normal & shear), we must calculate the stresses at different points for the critical section.

Loading Conditions
- $F = 0$
- $V_x = 0$
- $V_y = 28.5$ KN
- $M_x = 57$ KNm
- $M_y = 0$
- $M_z = 0$

* For every loading condition present, we must calculate an area property, as we will substitute both in the stress equation.

loading cond.		Area property
V_y	$\longrightarrow$	I_x , Q [At the required points]
M_x	$\longrightarrow$	I_x

3) Area properties:



- Cross section is symetrical around both axes x-x & y-y

⇒ centroid is at point "b" (at the centre of the section).

$$I_x = \frac{bh^3}{12} = \frac{150(250)^3}{12} = 195312500 \text{ mm}^4$$

$I_x = 195312500$.

In General ; $Q = \underbrace{A}_{\text{مساحة فوق أو تحت النقطة في حالة } V_y} \times \underbrace{L}_{\text{البعد بين centroid المساحة و centroid المساحة الكلية}}$

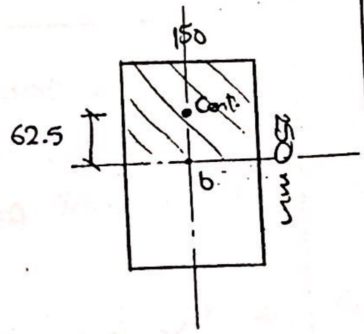
(أو يمين أو شمال النقطة في حالة V_x)

$Q_a = 0$

$Q_c = 0$

$Q_b = (150 \times 125) \times 62.5$

$= 1171875 \text{ mm}^3$



Loading conditions

$F = 0$

$V_x = 0$

$V_y = 28.5$ kN

$M_x = 57$ kNm

$M_y = 0$

$M_z = 0$

Area properties

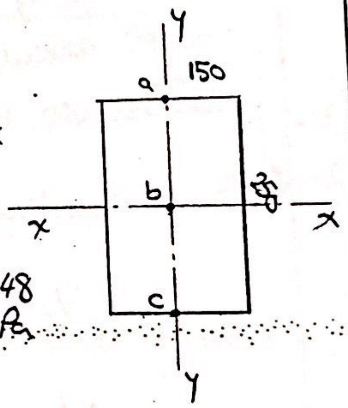
$I_x = 195312500 \text{ mm}^4$

$Q_a = 0$

$Q_b = 1171875 \text{ mm}^3$

$Q_c = 0$

4) Stress calculation :



(a) ; $\sigma = \pm \underbrace{\cancel{\frac{F}{A}}}_{=0} \pm \frac{M_x}{I_x} y \pm \underbrace{\cancel{\frac{M_y}{I_y}}}_{=0} x$

$\sigma_a = -\frac{57 \times 10^6}{195312500}(125) = -36.48$ MPa

$\sigma_b = 0 \rightsquigarrow$ because $y = 0$.

$\sigma_c = +36.48$ MPa

(b) $\tau = \frac{V_y Q}{I_x t}$ OR $\underbrace{\cancel{\frac{V_x Q}{I_y t}}}_{=0}$ (because $V_x = 0$)

$\tau_a = 0 \rightsquigarrow$ because $Q_a = 0$.

$\tau_b = \frac{(28.5 \times 10^3)(1171875)}{(195312500)(150)} = 1.14$ MPa

$\tau_c = 0 \rightsquigarrow$ because $Q_c = 0$.

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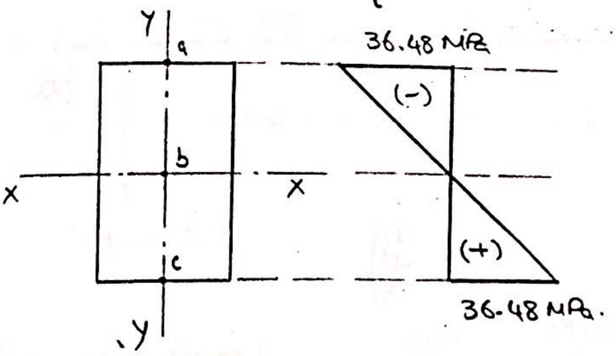
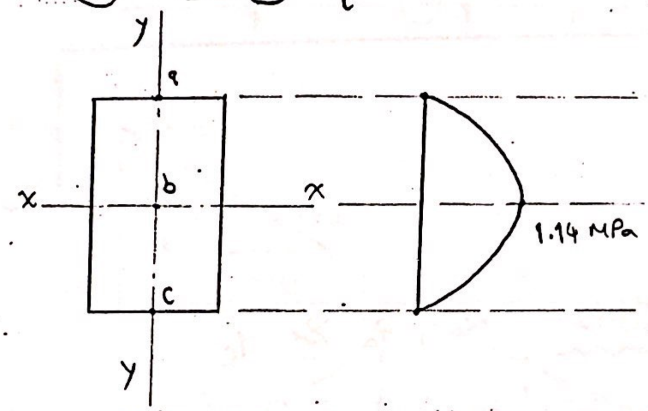
5) Stress diagrams :

ⓐ NSD [normal stress diagram].



ⓑ SSD [shear stress diagram].



6) Infinitesimal element :

$\sigma_a = -36.48$ MPa

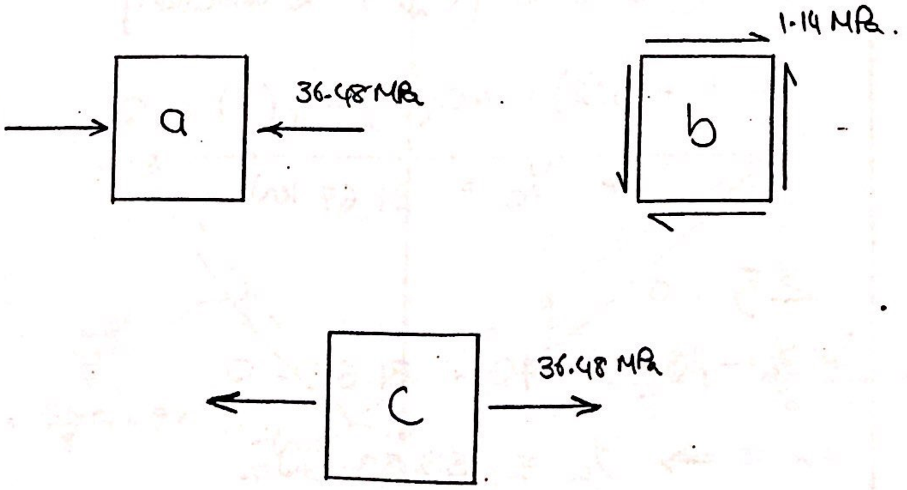
$\sigma_b = 0$

$\sigma_c = 36.48$ MPa

$\tau_a = 0$

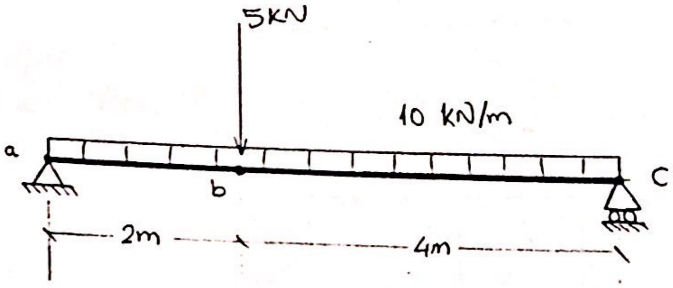
$\tau_b = 1.14$ MPa

$\tau_c = 0$

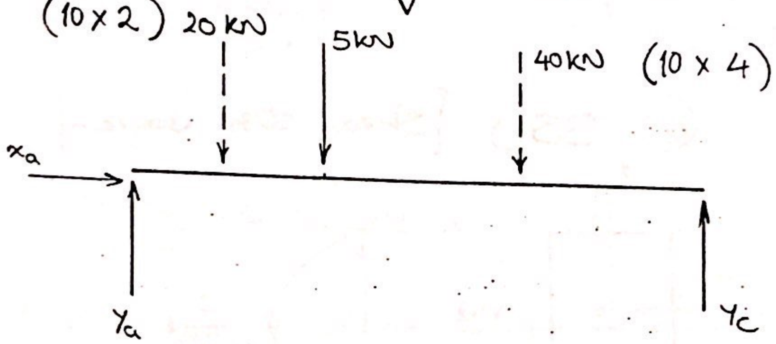


$\boxed{1}$

b]



$$\Downarrow$$



1) Reactions : x_a, Y_a & Y_c

$\sum F_x = 0$ [(→) +ve direction]

$x_a = 0$

$\sum M @ a = 0$ [(↻) +ve direction]

$20(1) + 5(2) + 40(4) - Y_c(6) = 0$

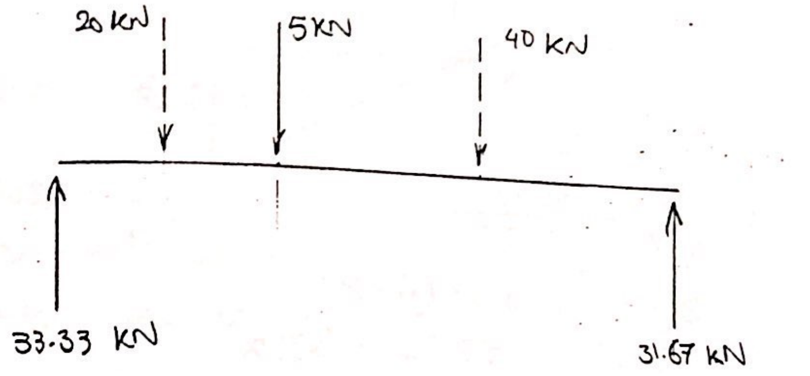
$\Rightarrow Y_c = 31.67$ kN.

$\sum F_y = 0$

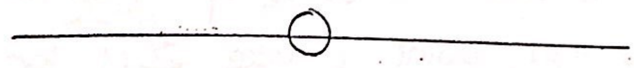
$Y_a - 20 - 5 - 40 + 31.67 = 0$

$\Rightarrow Y_a = 33.33$ kN

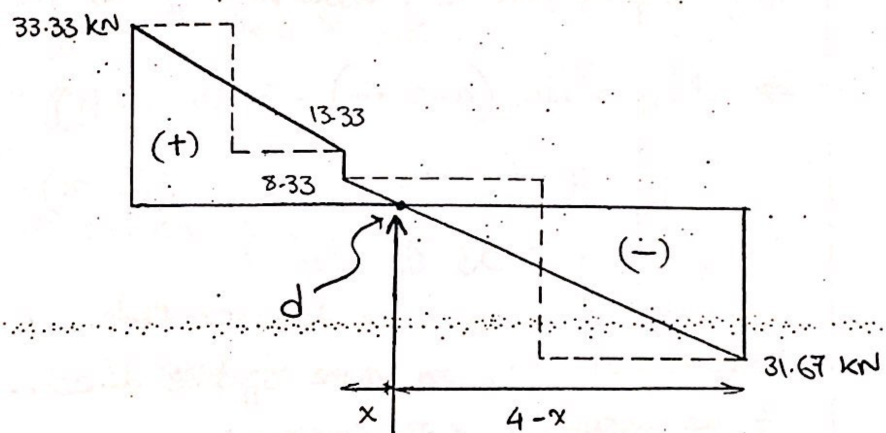
2) Loading diagrams :



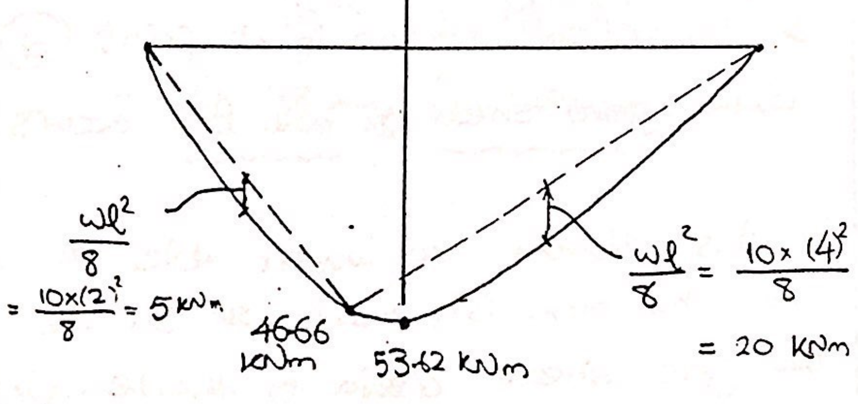
NFD



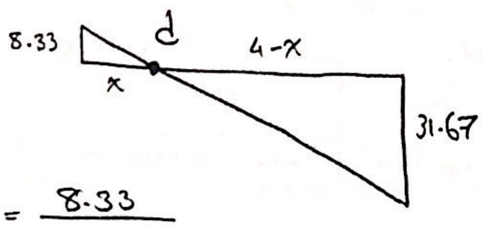
SFD



BMD



- From similar triangles :



$$\frac{x}{4-x} = \frac{8.33}{31.67}$$

$$31.67\,x = 8.33\,(4-x)$$

$$31.67\,x = 33.32 - 8.33\,x$$

$$\Rightarrow x = 0.833 \text{ m}$$

∴ To find the value of the bending moment at that point ; take ΣM for left or right of point d, assuming (↻) → +ve direc.

$$\Rightarrow M_d = 40\,(4-x-2) - 31.67\,(4-x)$$

$$= 40\,(4-0.833-2) - 31.67\,(4-x)$$

$$= (-)\,53.62 \text{ KNm}.$$

means that M_d is in the opposite direction of the assumed +ve direction.

⇒ maximum bending moment = 53.62

& the critical section is at point (d) where zero shear & max. BM occurs.

In this problem we weren't able to know where the max. BM occurs & so we used the zero shear criteria to decide. Unlike problem $\boxed{1}$

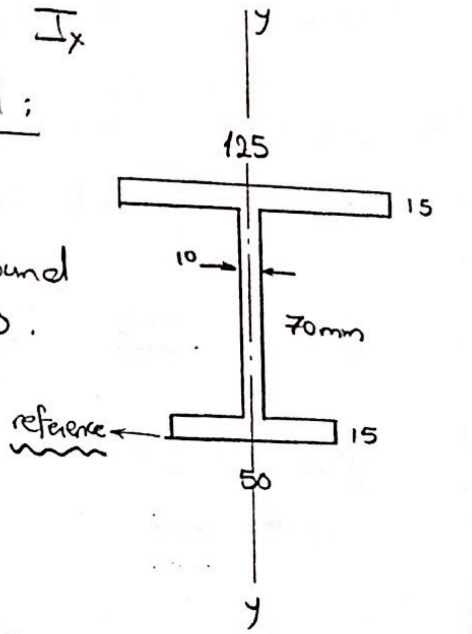
At critical section — loading conditions

$F = 0$
$V_x = 0$
$V_y = 0$
$M_x = 53.62$ KNm
$M_y = 0$
$M_z = 0$

3) Area properties ;

$M_x \longrightarrow I_x$

ⓐ Position of centroid ;

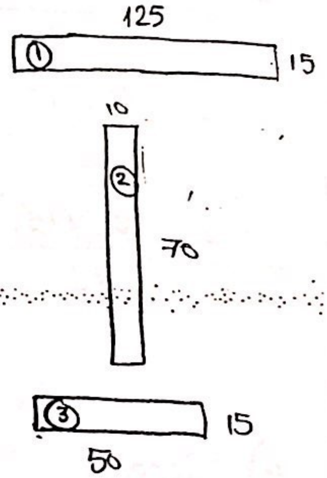


∵ cross section is symmetrical around y-axis $\Rightarrow \bar{x} = 0$.

&

$$\bar{y} = \frac{\Sigma A_i Y_i}{\Sigma A_i}$$

we will divide the cross section into regular shapes & take a reference



A_i : Area of each part.

Y_i : distance between part i centroid & reference.

$$\Rightarrow \bar{y} = \frac{(125\times15)(92.5)+(10\times70)(50)+(50\times15)(7.5)}{(125\times15)+(10\times70)+(50\times15)}$$

$= 64.38$ mm from reference.

$\bar{y} = 64.88$ mm

To find I_x, we will use the parallel axis theorem (نظرية نقل المحاور المتوازية)

The cross section is already divided into 3 shapes on which we will apply the theorem

$$I_x = \sum \left[I_{x'} + A\left(\overline{xx'}\right)^2 \right]$$

"I" لكل شكل حول محوره

مساحة الشكل

المسافة بين محور الشكل و محور القطاع كله

$$\Rightarrow I_x = \frac{(125)(15)^3}{12} + \left[(125\times15)(92.5-64.38)^2\right]$$
$$+ \frac{(10)(70)^3}{12} + \left[(10\times70)(64.38-50)^2\right]$$
$$+ \frac{(50)(15)^3}{12} + \left[(50\times15)(64.38-7.5)^2\right]$$

$$= 4388928.96 \text{ mm}^4$$

$I_x =$ 438892

4) Stress calculation :

$$\sigma = \pm \underbrace{\cancel{\frac{F}{A}}}_{=0} \pm \frac{M_x}{I_x} y \pm \underbrace{\cancel{\frac{M_y}{I_y}}}_{=0} x$$

$$\sigma_a = -\frac{53.62\times10^6}{4388928.96}(100-64.38) = -435.2 \text{ MPa}$$

$\sigma_a = -43$

$$\sigma_b = \frac{-53.62\times10^6}{4388928.96}(85-64.38) = -251.92 \text{ MPa}$$

$\sigma_b = -25$

$\sigma_c = 0 \rightsquigarrow$ as $y = 0$.

$$\sigma_d = + \frac{53.62 \times 10^6}{4388928.96} (64.38 - 15) = 603.3 \text{ MPa}$$

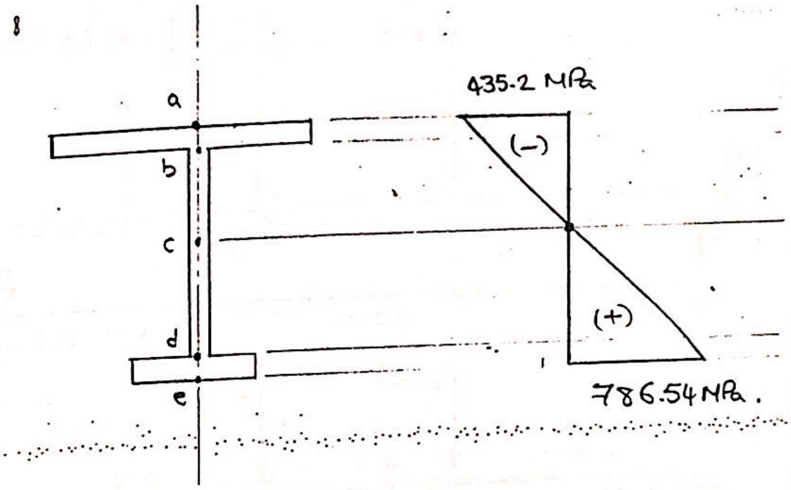
$$\sigma_e = + \frac{53.62 \times 10^6}{4388928.96} (65.38) = 786.54 \text{ MPa}.$$

$\sigma_c = 0$

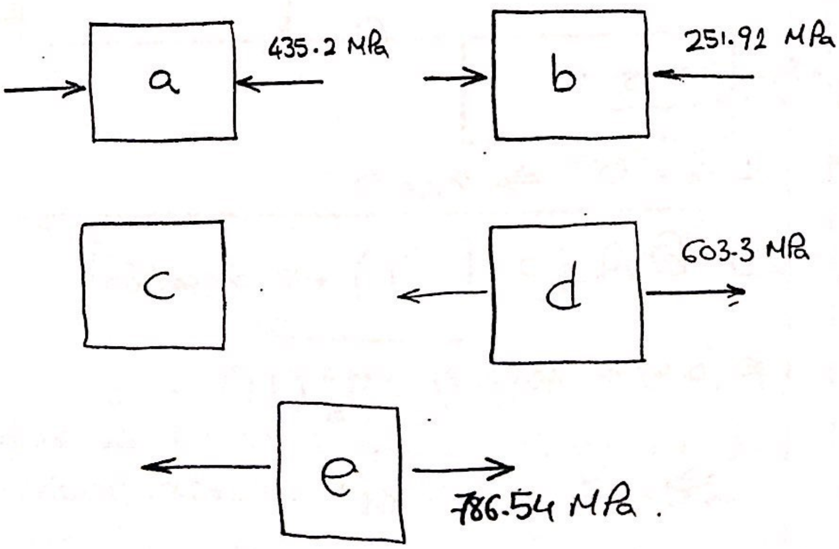
$\sigma_d = 603.3$ M

$\sigma_e = 786.54$ M

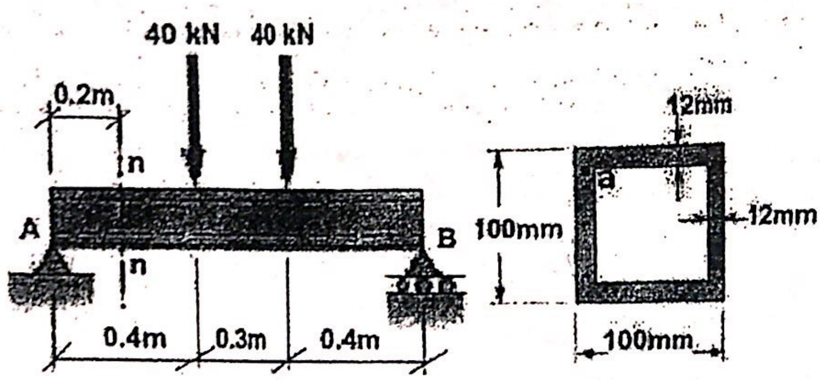
5) Stress diagrams :

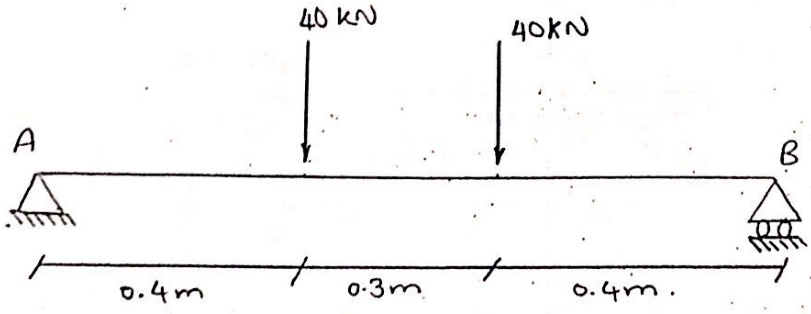


6) Infinitesimal elements :

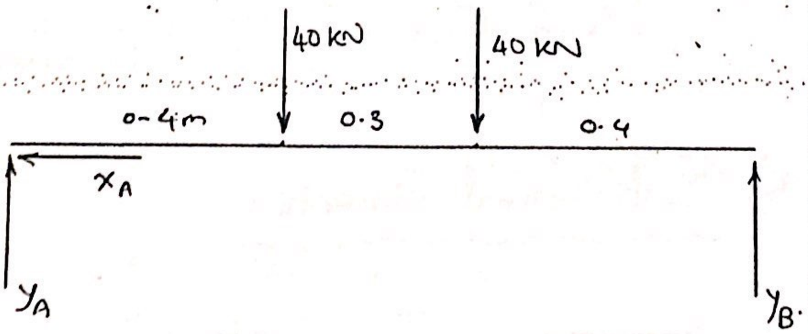


1 c





FBD



*Reactions :

$\sum F_x = 0 \Rightarrow x_A = 0$ $\qquad x_A = 0$.

$\sum M @ A = 0$ [(↻) +ve direction].

$40(0.4) + 40(0.7) - y_B(1.1) = 0$.

$\Rightarrow y_B = 40$ kN $\qquad y_B = 40$ kN

$$\sum F_y = 0$$

$$40 - 40 - 40 + Y_B = 0.$$

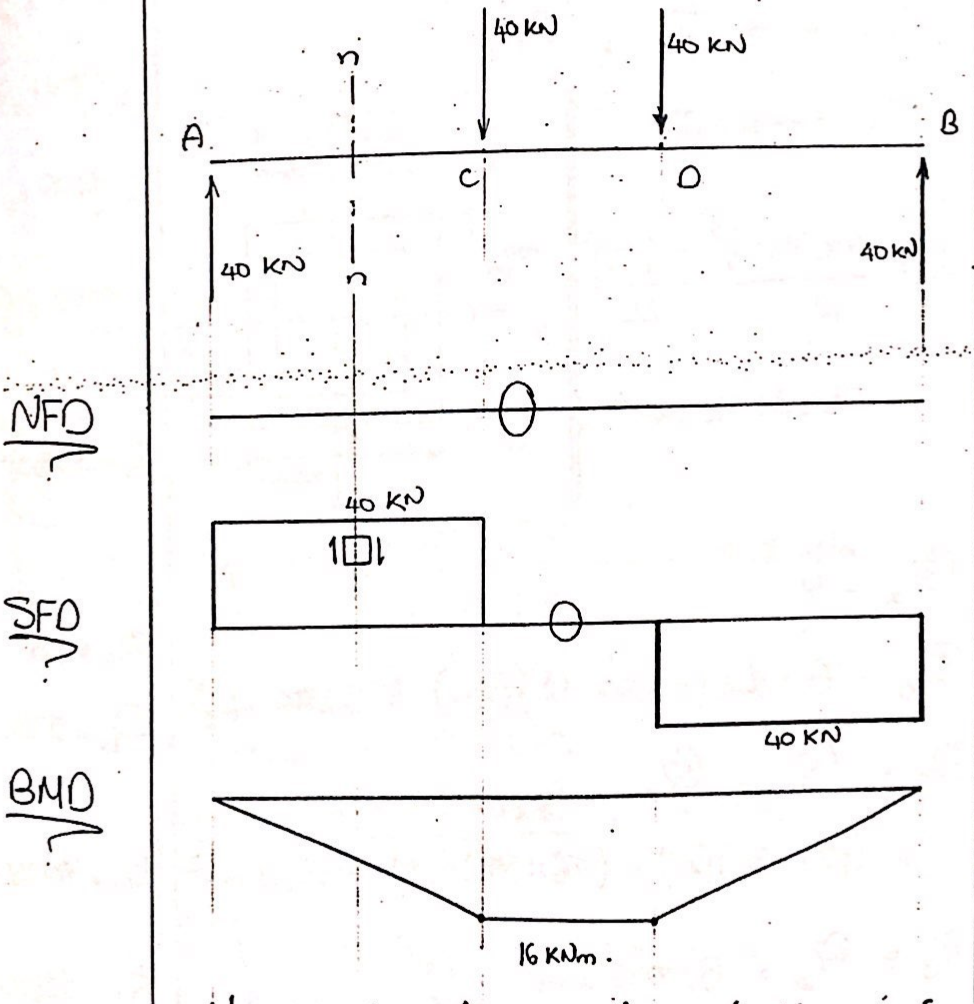
$$\Rightarrow Y_B = 40\ kN$$

$Y_B = 40\ kN.$

ملحوظة

* إذا كان الـ beam متماثل حول خط يمر بمنتصف الـ beam

⇐ قيمة كل reaction يساوي محصلة الأحمال مقسوم على "2" وعكس إتجاه المحصلة



الملحوظة:

The critical section is at "C" just lef
and
at "D" just rig

- في هذه المسألة مطلوب مني أشتغل عند قطاع محدد وهو n-n وليس عند الـ critical section

At sec n-n :

*** loading conditions**

$F = 0$	$M_x = 40 \times 0.2 = 8\ kNm$
$V_x = 0$	$M_y = 0$
$V_y = 40\ kN$	$M_z = 0$.

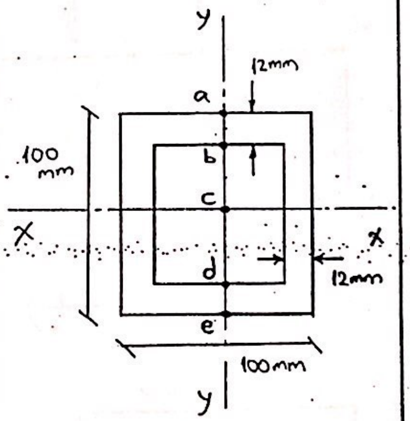
*** Area properties :**

$V_y \longrightarrow I_x\ ,\ Q$

$M_x \longrightarrow I_x$

$$I_x = \frac{(100)(100)^3}{12} - \frac{(76)(76)^3}{12}$$

$$= 5553152\ mm^4$$



$Q_a = 0$

$Q_b = A \times L = (100 \times 12)(44) = 52800\ mm^3$

$Q_c = Q_1 - Q_2$

$= (100 \times 50)(25) - (76 \times 38)(19) = 70128\ mm^3$

$Q_d = Q_b = 52800\ mm^3$

$Q_e = 0$.

* stress calculations:

loading conditions

$V_y = 40\ kN$

$M_x = 8\ kN m$

Area properties

$I_x = 5553152\ mm^4$

$Q_a = 0$

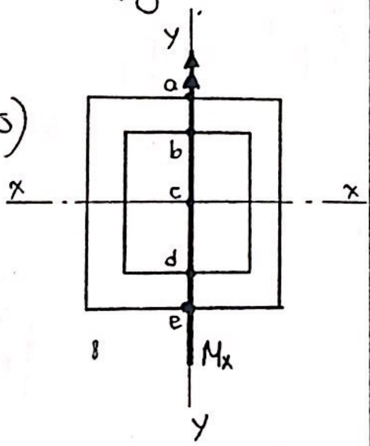
$Q_b = 52800\ mm^3$

$Q_c = 70128\ mm^3$

$Q_d = 52800\ mm^3$

$Q_e = 0$

$$\sigma = \pm \underbrace{\frac{F}{A}}_{0} \pm \frac{M_x}{I_x} y \pm \underbrace{\frac{M_y}{I_y}}_{0} x$$



$$\sigma_a = -\frac{8 \times 10^6}{5553152}(50)$$

$$= -72\ MPa$$

$$\sigma_c = 0 \rightsquigarrow \left[\begin{array}{c}\text{because} \\ y = 0\end{array}\right]$$

$$\sigma_e = +\frac{8 \times 10^6}{5553152}(50) = 72\ MPa$$

$$\tau = \frac{V_y Q}{I_x t}$$

$$\tau_a = 0 \rightsquigarrow Q_a = 0.$$

$$\tau_b\Big|_{above} = \frac{(40 \times 10^3)(52800)}{(5553152)(100)} = 3.8\ MPa.$$

$$\tau_b\Big|_{below} = \frac{(40 \times 10^3)(52800)}{(5553152)(24)} = 15.8\ MPa.$$

$$\tau_c = \frac{(40 \times 10^3)(70128)}{(5553152)(24)} = 21.\ MPa.$$

$$\tau_d\Big|_{above} = \tau_b\Big|_{below} = 15.8\ MPa.$$

$$\tau_d\Big|_{below} = \tau_b\Big|_{above} = 3.8\ MPa \quad \Big\} \quad \tau_e = 0.$$

$\sigma_a = -72\ MPa$

$\sigma_c = 0$

$\sigma_e = 72\ MPa$

$\tau_a = 0$

$\tau_b\Big|_{above} = 3.8\ MPa$

$\tau_b\Big|_{below} = 15.8\ MPa$

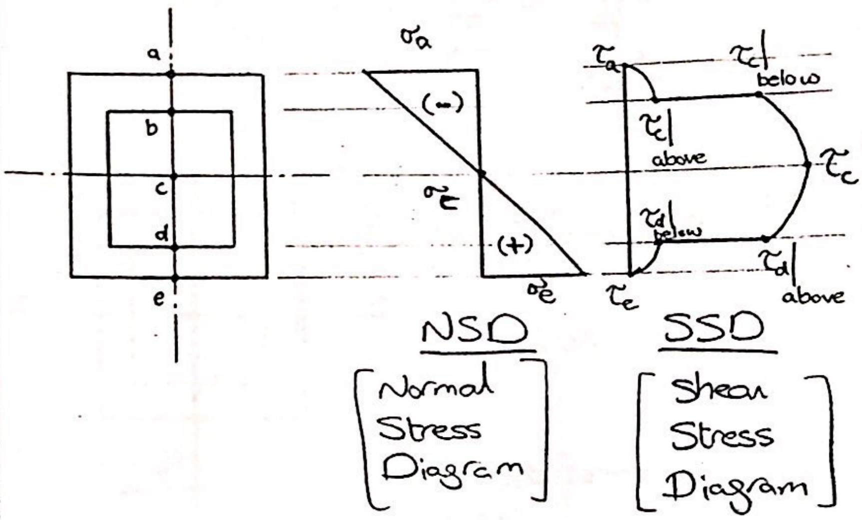
$\tau_c = 21\ MPa$

$\tau_d\Big|_{above} = 15.8\ MPa$

$\tau_d\Big|_{below} = 3.8\ MPa$

$\tau_e = 0.$

* Stress diagrams :



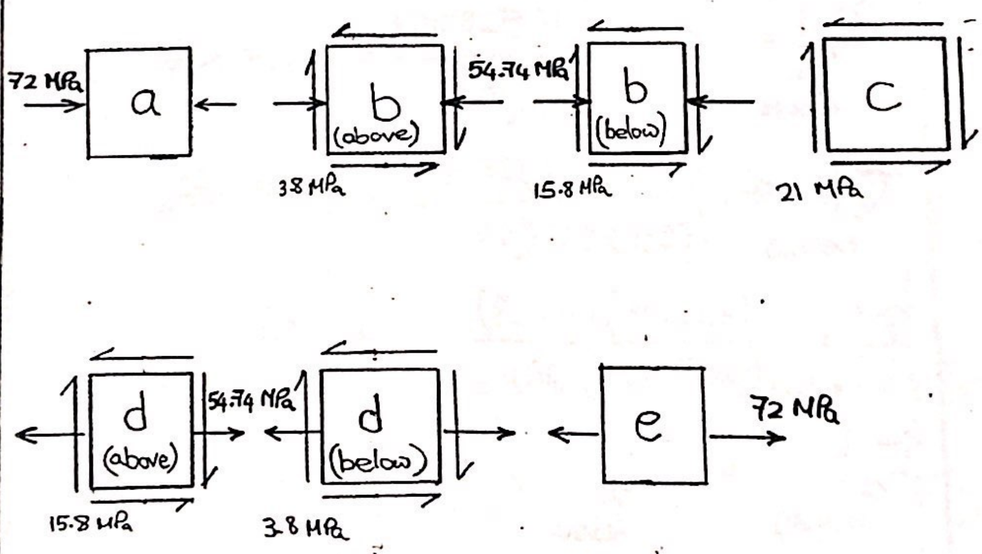
$$\sigma_b = -\frac{8\times10^6}{5553152}(38) = -54.74\ MPa.$$

$$\sigma_d = +54.74\ MPa.$$

$\sigma_a = -72$ MPa
$\sigma_b = -54.74$ MPa
$\sigma_c = 0$
$\sigma_d = +54.74$ MPa
$\sigma_e = 72$ MPa

$\tau_a = 0.$
$\tau_b|_{ab} = 3.8$ MPa
$\tau_b|_{bel} = 15.8$ MPa
$\tau_c = 21$ MPa
$\tau_d|_{ab} = 15.8$ MPa
$\tau_d|_{bel} = 3.8$ MPa
$\tau_e = 0$

* Infinitesimal elements :



Problem 2

(FIGURE 1)

The shown frame is subjected to the given loads and has the shown cross section .
a) Calculate and draw the normal and shear stress diagrams at section C-C.
b) Determine the state of stresses at points G and H showing the results on an infinitesimal elements.

$F_1 = 5$ kN & $F_2 = 8$ kN

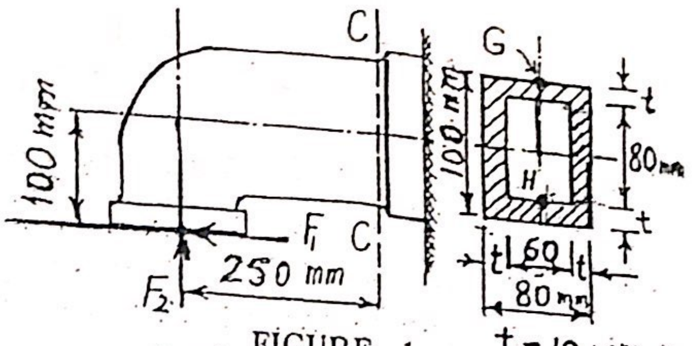


FIGURE 1

$t = 10$ mm

Loading conditions :

$F = +5\text{ kN} = F_1$

$V_x = 0$

$V_y = 8\text{ kN} \uparrow$

$M_x = (5 \times 100) + (8 \times 250) = 2500\text{ kN mm} \Uparrow$

$M_y = 0$

$M_z = 0$

$F = +5$ kN

$V_y = 8$ kN $\uparrow$

$M_x = 2500$ kN mm $\Uparrow$

Area properties :

$F \rightarrow A$

$V_y \rightarrow I_x\ ,\ Q_G\ \&\ Q_H$

$M_x \rightarrow I_x$

$A = (100 \times 80) - (80 \times 60) = 3200\text{ mm}^2$

$I_x = \frac{(80)(100)^3}{12} - \frac{(60)(80)^3}{12} = 4106666.667\text{ mm}^4$

$A = 3200\text{ mm}^2$

$I_x = 4106666.667\text{ mm}^4$

$Q_G = 0$

$Q_H = (80 \times 10)(40+5) = 36000 \text{ mm}^3$

Stress Calculations :

ⓐ Normal stresses

$$\sigma = \pm \frac{F}{A} \pm \frac{M_x}{I_x} y \pm \cancel{\frac{M_y}{I_y}} x$$

0

$$\sigma_G = + \frac{5 \times 10^3}{3200} - \frac{2500 \times 10^3}{4106666.667}(50)$$

$$\sigma_H = + \frac{5 \times 10^3}{3200} + \frac{2500 \times 10^3}{4106666.667}(40)$$

ⓑ Shear stresses :

$$\tau = \frac{V_y Q}{I_x t}$$

$\tau_G = 0$

$$\tau_H\Big|_{\text{above}} = \frac{(8 \times 10^3)(36000)}{(4106666.667)(20)} = 3.5 \text{ MPa}$$

$$\tau_H\Big|_{\text{below}} = \frac{(8 \times 10^3)(36000)}{(4106666.667)(80)} = 0.87 \text{ MPa}$$

$Q_G = 0$

$Q_H = 36000$ m

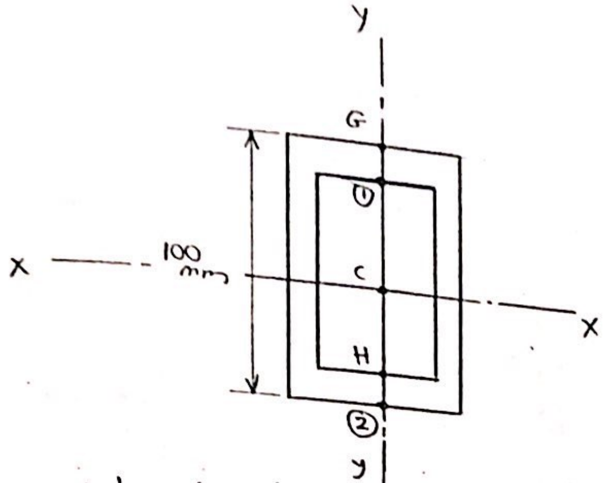
$\sigma_G = -28.87$ MPa

$\sigma_H = 25.91$

$\tau_G = 0$

$\tau_H\Big|_{\text{above}} = 3.5$

$\tau_H\Big|_{\text{below}} = 0.8$



Extra points to draw stress diagrams:-

$$\sigma_2 = +\frac{F}{A} + \frac{M_x}{J_x} y$$

$$= +\frac{5\times10^3}{3200} + \frac{2500\times10^3}{4106666.667}(50) = 32\ \text{MPa}$$

$\sigma_2 = 32\ \text{MPa}$

same as Q_H

$$\tau_1\Big|_{\text{above}} = \frac{(8\times10^3)(36000)}{(4106666.667)(80)} = 0.87\ \text{MPa} = \tau_H\Big|_{\text{above}}$$

$\tau_1\Big|_{\text{above}} = 0.87\ \text{MPa}$

$$\tau_1\Big|_{\text{below}} = \frac{(8\times10^3)(36000)}{(4106666.667)(20)} = 3.5\ \text{MPa} = \tau_H\Big|_{\text{below}}$$

$\tau_1\Big|_{\text{below}} = 3.5\ \text{MPa}$

$$-\ Q_C = (50\times80)(25) - (60\times40)(20)$$

$$= 52000\ \text{mm}^3$$

$Q_C = 52000\ \text{mm}^3$

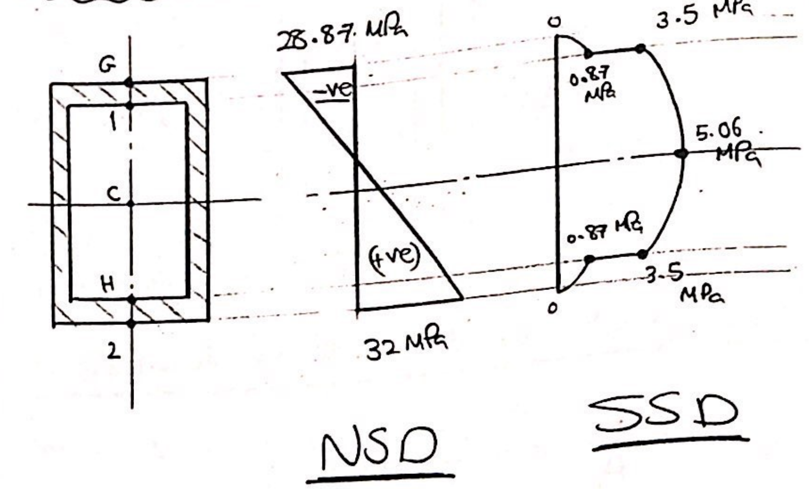
$$\Rightarrow \tau_C = \frac{(8\times10^3)(52000)}{(4106666.667)(20)} = 5.065\ \text{MPa}$$

$\tau_c = 5.065\ \text{MPa}$

$$\tau_2 = 0$$

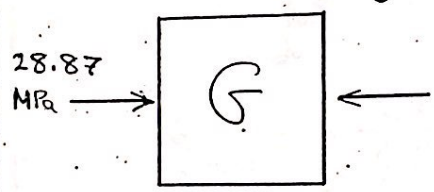
$\tau_2 = 0.$

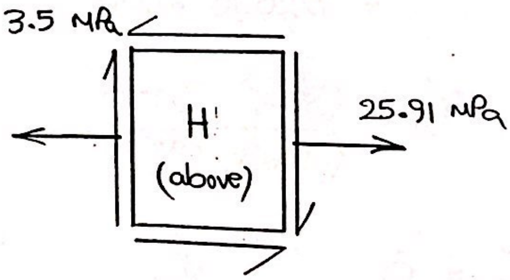
Stress diagrams :

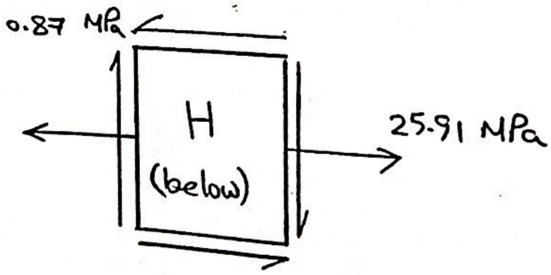


Infinitesimal elements :

$\tau_G = 0.$







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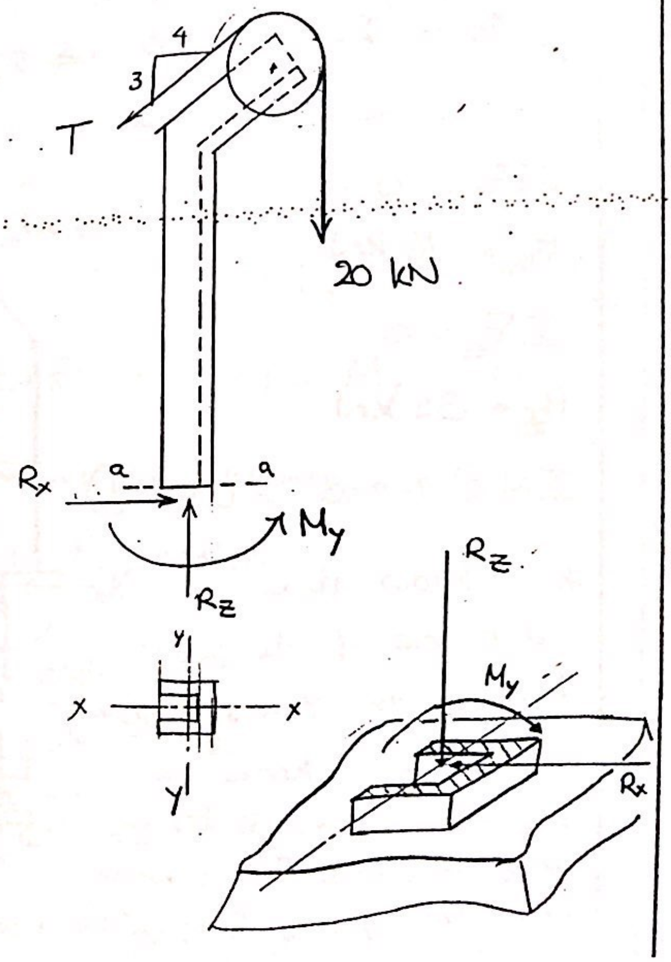
Problem 3

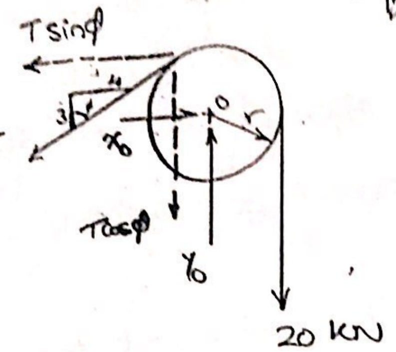
Same as problem 2, we don't need to look for the critical section. It is required to calculate the stresses at a specified section

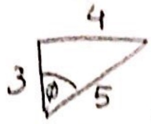
We will start by finding the loading conditions at the required section (sec. a-a).

ومعنى إيجاد الـ loading cond. عند قطاع معين هو إيجاد الـ internal forces عند هذا القطاع.



ΣM @ point O = 0





$\cos\phi = 3/$

$\sin\phi = 4/$

$T\cancel{r} = 20\cancel{r}$

$\Rightarrow T = 20\,kN$

$\Sigma F_y = 0$

$y_0 - T\cos\phi - 20 = 0$

$y_0 - \cancel{20}^{4}(3/\cancel{5}_{1}) - 20 = 0$

$y_0 - 12 - 20 = 0 \Rightarrow y_0 = 32\,kN$

$y_0 = 32$

$\Sigma F_x = 0$

$x_0 - T\sin\phi = 0$

$x_0 - \cancel{20}^{4}(4/\cancel{5}_{1}) = 0 \Rightarrow x_0 = 16\,kN$

$x_0 = 16\,k$

$\Sigma F_x = 0$

$R_x = 16\,kN$

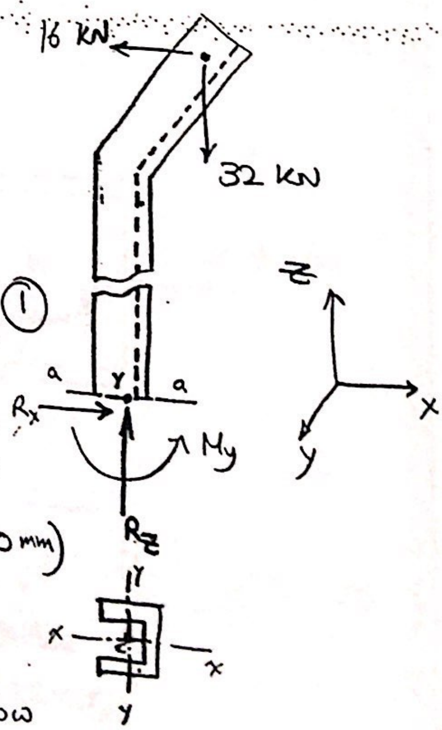
$R_x = 16\,k$

$\Sigma F_z = 0$

$R_z = 32\,kN$

$R_z = 32$

ΣM @ y-axis $= 0 \rightarrow$ 1



* I know the distance of the 16 kN force to the y-axis (1800 mm) but I don't know the distance of the 32 kN force because I don't know where the y-axis falls on the horizontal plane.

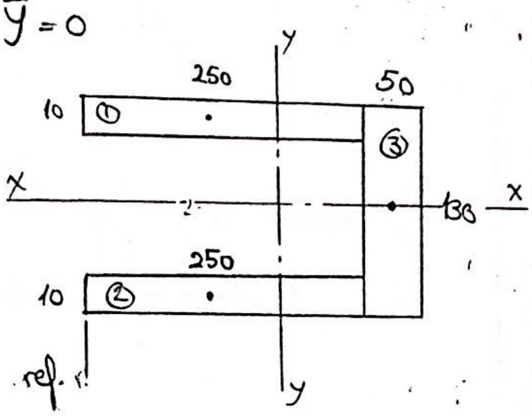
the y-axis passes through the centroid of the cross section area & therefore we must calculate the position of the centroid.

$\because$ Area is symmetrical around x-x

$\Rightarrow$ lies on it $\Rightarrow \bar{y} = 0$

But ;

$$\bar{x} = \frac{\sum A_i L_i}{\sum A_i}$$



$$= \frac{2(250\times10)(125) + (50\times130)(275)}{2(250\times10) + (50\times130)} = 209.8\ mm$$ from ref.

$\bar{x} = 209.8\ m$

Back to equation ① ;

$\sum M$ @ y-axis $= 0$ [ass. ↻ +ve direction].

$$32(600 - 209.8) - 16(1800) - M_y = 0$$

$$\Rightarrow M_y = -16313.6\ kNmm$$

$$= -16.3\ kNm$$

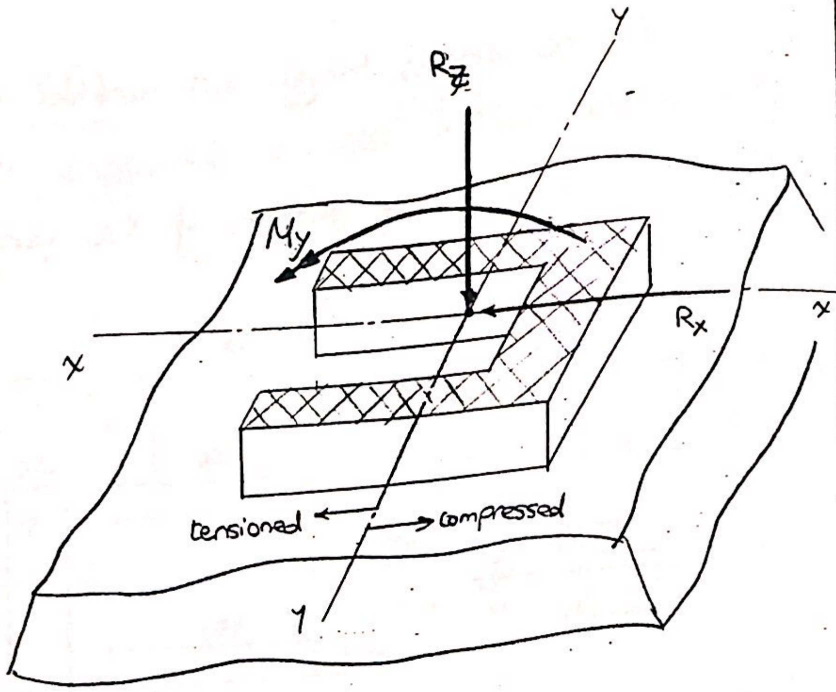
$M_y = 16.3\ kN$

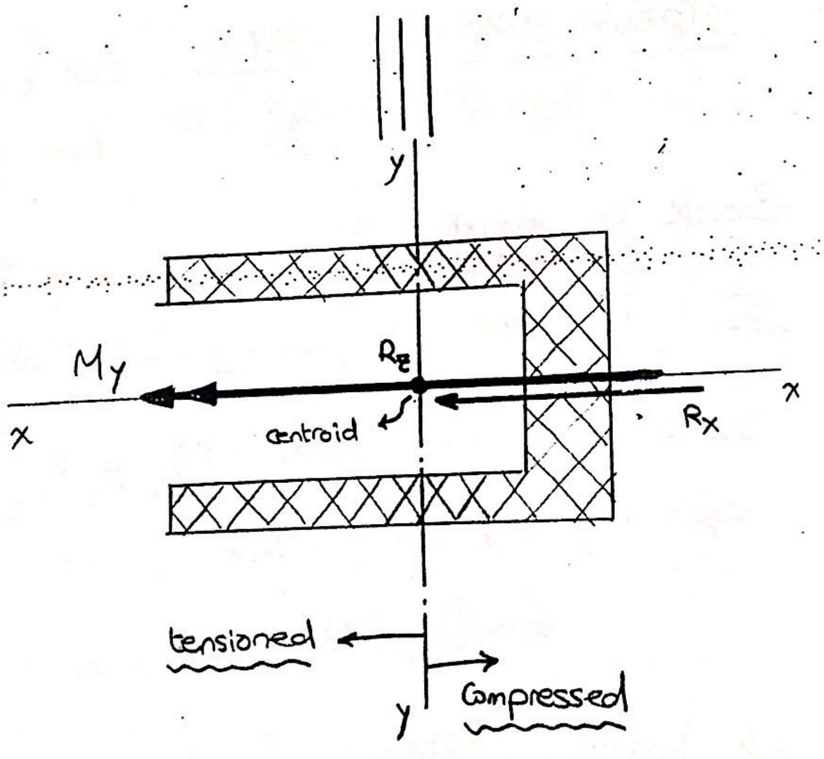
$\Rightarrow$ Loading conditions ;

$R_x = 16\ kN \equiv V_x$

$R_z = 32\ kN \equiv F$

$M_y = 16.3\ kNm\ \times$





:لو النقطة مع رأس السهم ⇐ (-ve)

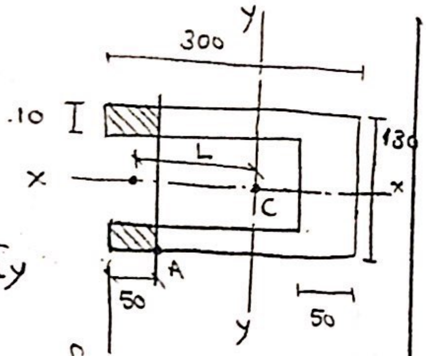
ولو مع ذيل السهم ⇐ (+ve)

Area properties:

$F \longrightarrow A$

$V_x \longrightarrow Q, I_y$

$M_y \longrightarrow I_y$



refer.

$$A = (300 \times 130) - (250 \times 110) = 11500 \text{ mm}^2$$

$A = 11500$

$$Q_A = A \times L$$

(الجزء دي المساحة المختارة يمين أو شمال النقطة)

$$= \left[(50 \times 130) - (50 \times 110)\right] \times (209.8 - 25)$$

$$= 184800 \text{ mm}^3$$

$Q_A = 184800$ mm

$$I_y = \frac{(130)(300)^3}{12} + (300 \times 130)(209.8 - 150)^2$$

$$- \frac{(110)(250)^3}{12} + (250 \times 110)(209.8 - 125)^2$$

$$= 90982793.33 \text{ mm}^4$$

$I_y = 90982793.33$ mm^4

at sec a-a loading conditions

$F = 32\ kN$

$V_x = 16\ kN$

$V_y = 0$

$M_x = 0$

$M_y = 16.3\ kNm$

$M_z = 0$

Area properties

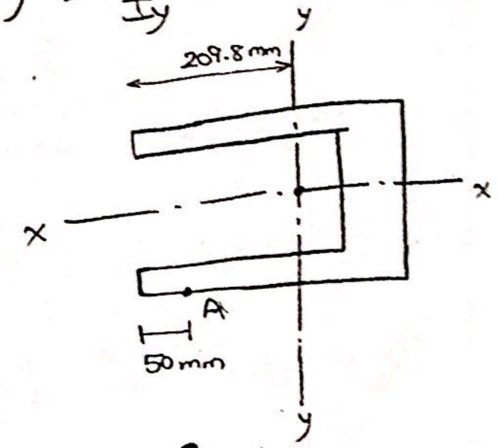
$A = 11500\ mm^2$

$Q_A = 184800\ mm^3$

$I_y = 90982793.33\ mm^4$

Stress Calculations :

$$\sigma = \pm \frac{F}{A} \pm \underbrace{\frac{M_x}{I_x} y}_{0} \pm \frac{M_y}{I_y} x$$



$$\Rightarrow \sigma_A = -\frac{32 \times 10^3}{11500} - \frac{16.3 \times 10^6}{90982793.33}(209.8 - 50)$$

$$= \boxed{-31.4\ MPa}$$

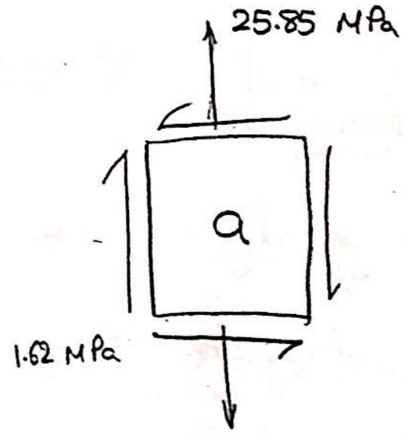
$\sigma_a = 25.$

$$\tau = \frac{V_x Q}{I_y t}$$

$$\tau_a = \frac{(16 \times 10^3)(184800)}{(90982793.33)(20)} = \boxed{1.62\ MPa}$$

$\tau_a = 1.6$

Infinitesimal element

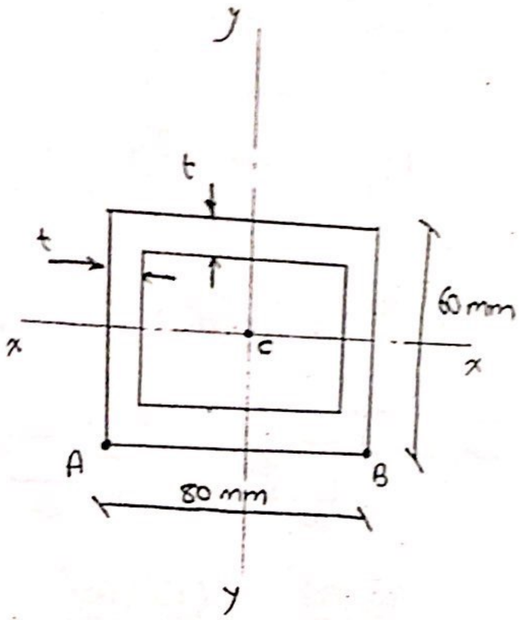


Problem 4

given :

t = 10 mm

P = 30 KN



* أولاً أقوم بنقل القوى إلى centroid القطاع "C" اللي أنا عايز أحسب عنده، ثم أستنتج الـ loading conditions

ملحوظة

الجملة السابقة هتديني نفس نتيجة لو قطعت و وجدت الـ internal forces

* Loading conditions :

$F = P = 30\ KN$

$V_x = 0$

$V_y = 0$

$M_x = 0$

$M_y = P \times (0.2 + 0.04)$

$= 30 \times 0.24 = 7.2\ kNm$

$M_z = 0$.

$F = 30\ KN$

$M_y = 7.2\ KNm$

Loading conditions

F= 30 KN

$M_y = 7.2 kNm$

* Area properties :

$F \longrightarrow A$

$M_y \longrightarrow I_y$

$A = (80\times60)-(60\times40) = 2400\ mm^2$

A= 2400

هنالا يوجد
نقل محاور
يلزم .cent كل الأشكال
متطبعة على cent. القطاع كامل

$I_y = I_{\square\ 80\times60} - I_{\square\ 60\times40}$

$= \frac{(60)(80)^3}{12} - \frac{(40)(60)^3}{12} = 1840000\ mm^4$

I_y = 184000

* Stress Calculations :

$\sigma = \pm\frac{F}{A} \pm \underbrace{\frac{M_x}{I_x}y}_{=0} \pm \frac{M_y}{I_y}x$

$\sigma_A = +\frac{30\times10^3}{2400} + \frac{7.2\times10^6}{1840000}(40) = 169\ MPa$

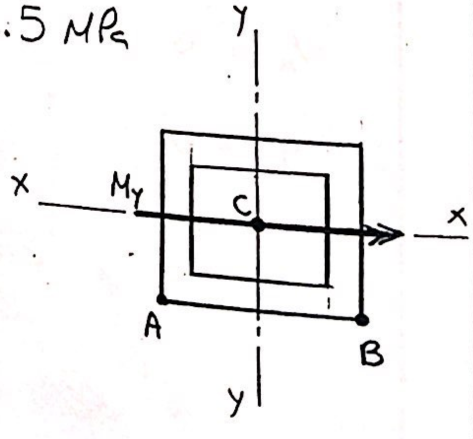
$\sigma_A = 169$

$\sigma_B = +\frac{30\times10^3}{2400} - \frac{7.2\times10^6}{1840000}(40) = -144\ MPa$

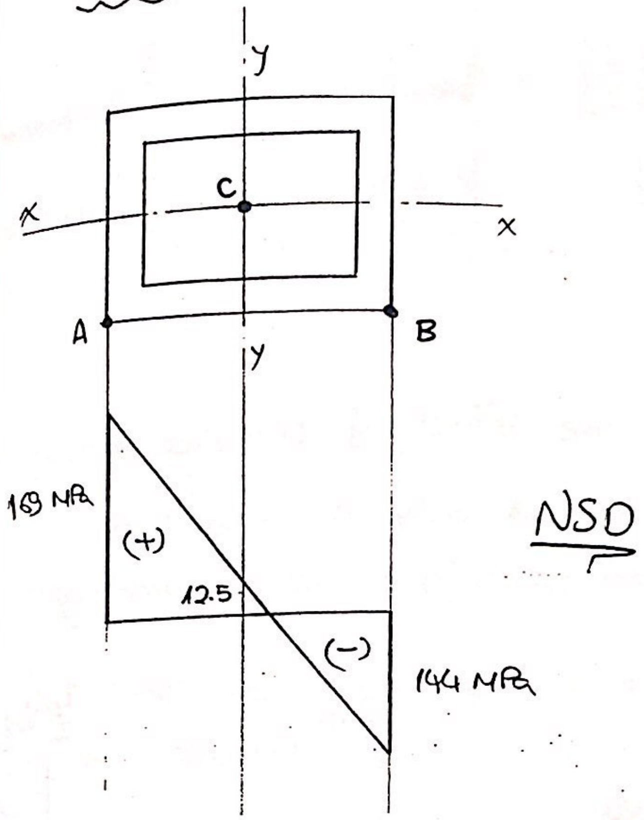
$\sigma_B = -144$

$\sigma_C = +\frac{30\times10^3}{2400} = 12.5\ MPa$

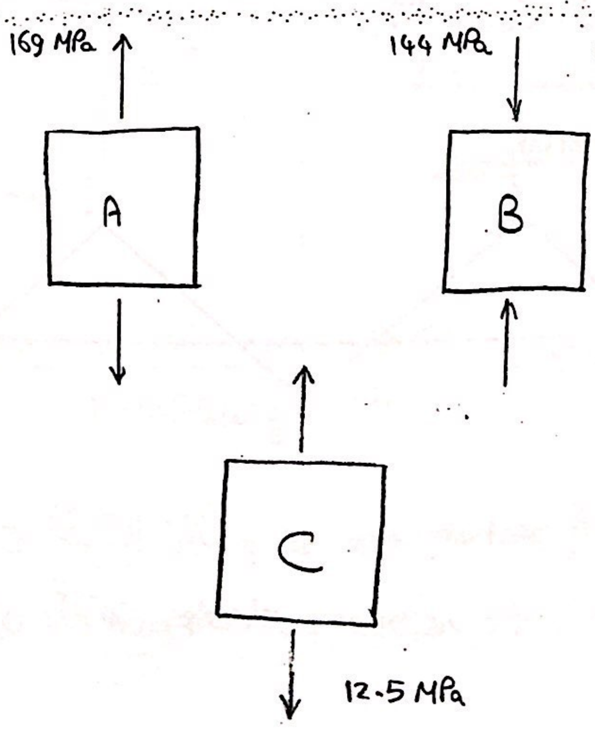
$\sigma_C = 12.5$ MPa



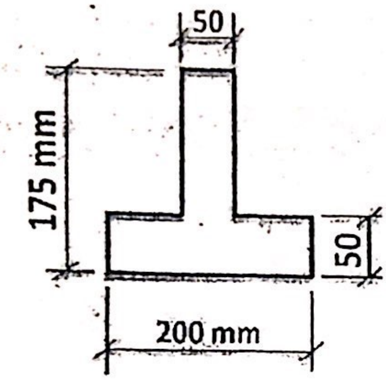
* Stress diagrams

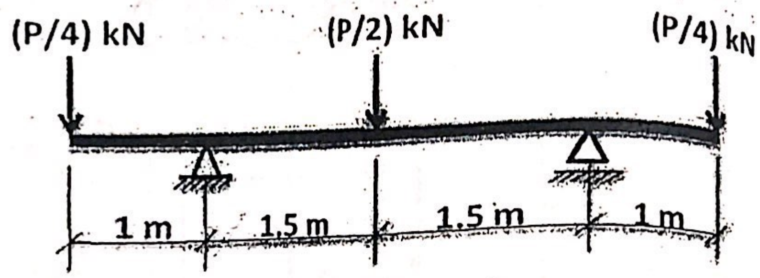


* infinesimal elements



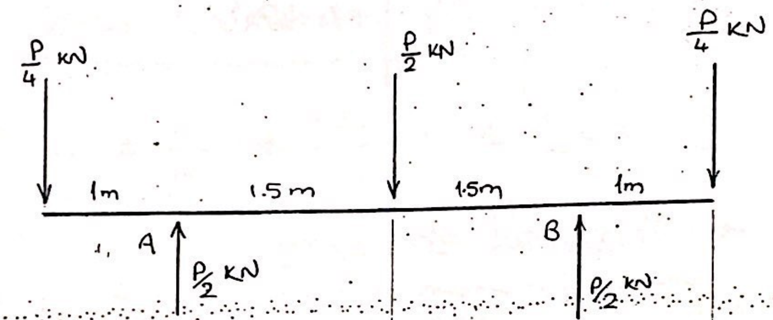
5



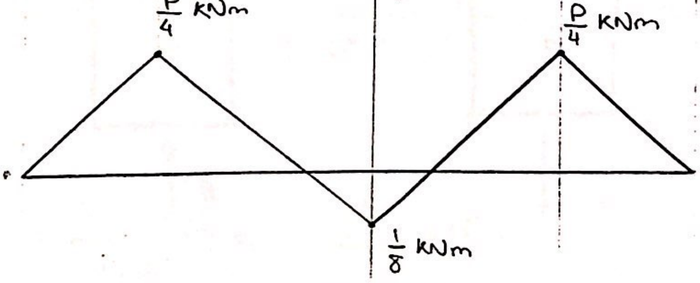


- Neglecting the effect of transverse shear

⇒ the beam will only be subjected to normal stress (σ) due to bending moment.



BMD



⇒ Critical section are at points A or B

يعني أقصى stresses موجودين عند النقط A or B

⇛ Loading conditions [Neglecting Transverse shear]

$F = 0$

V_x = neglected

V_y = neglected

$M_x = \frac{P}{4}$ kNm

$M_y = 0$

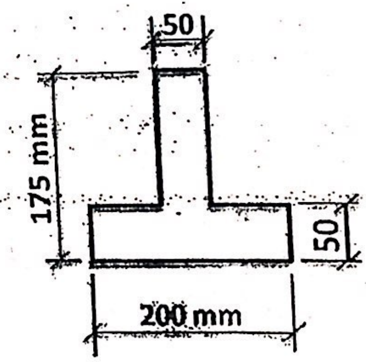
$M_z = T = 0$

$M_x = \frac{P}{4}$ kNm

* Area properties :

$M_x \longrightarrow I_x$

But first we need to know the position of the centroid ;



$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i}$$

$$= \frac{(200)(50)(25) + (50)(125)(112.5)}{(200)(50) + (50)(125)} = 58.7 \text{ mm}$$

$\bar{y} = 58.7$ mm

$$I_x = \frac{(200)(50)^3}{12} + (200)(50)(58.7 - 25)^2 + \frac{(50)(125)^3}{12} + (50)(125)(112.5 - 58.7)^2$$

$$= 39668504.2 \text{ mm}^4$$

$I_x = 39668504.2$ mm^4

loading conditions

$M = \frac{P}{4}$ KNm

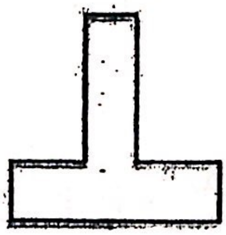
Area properties

$I_x = 39668504.2$ mm^4

Stress calculation :

$$\sigma = \pm \underbrace{\cancel{\frac{F}{A}}}_{=0} \pm \frac{M_x}{I_x} y \pm \underbrace{\cancel{\frac{M_y}{I_y}}}_{=0} x$$

* max normal stresses occur at a cross section at the furthest points from the neutral axis.



$$\sigma_1 = + \frac{P/4 \times 10^6}{39668504.2} (175 - 58.7) = 0.73 P \text{ Mpa}$$

$$\sigma_2 = - \frac{P/4 \times 10^6}{39668504.2} (58.7) = -0.37 P \text{ Mpa}$$

σ_1 is the maximum normal stress (type: tension) occuring on the cross section.

σ_2 is the maximum normal stress (type: compression) occurring on the cross section.

$\Rightarrow \sigma_1 \leqslant 35$ MPa & $\sigma_2 \leqslant 150$ MPa.

$0.73P \leqslant 35$ $\qquad$ $0.37P \leqslant 150$

$\Rightarrow P_{all} = 47.9$ KN $\qquad$ $\Rightarrow P_{all.} = 405.4$ KN

We select the smaller P to insure safety $\Rightarrow P_{all} = 47.9$ KN

$P_{all} = 47.9$ KNm

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